

45-mavzu. R^m fazo. R^m fazoda ochiq va yopiq to'plamlar

45-ma'ruza

Reja

- 1^o. R^m fazo tushunchasi.
- 2^o. R^m fazoda nuqtaning atrofi.
- 3^o. R^m fazoda ochiq va yopiq to'plamlar.

1^o. R^m fazo tushunchasi.

Haqiqiy sonlar to'plami R yordamida ushbu

$$\underbrace{R \times R \times \dots \times R}_m = \{ (x_1, x_2, \dots, x_m) : x_1 \in R, x_2 \in R, \dots, x_m \in R \} \quad (1)$$

to'plamni (R ning dekart ko'paytmalaridan tuzilgan to'plamni) hosil qilaylik.

Ravshanki, (1) to'plamning har bir elementi m ta $x_1, x_2, \dots, x_m$ haqiqiy sonlardan tashkil topgan tartiblangan m lik

$$(x_1, x_2, \dots, x_m)$$

dan iborat bo'ladi. Uni (1) to'plamning nuqtasi deyilib, bitta harf bilan belgilanadi:

$$x = (x_1, x_2, \dots, x_m).$$

Bunda $x_1, x_2, \dots, x_m$ sonlar x nuqtaning mos ravishda birinchi, ikkinchi, ... , m -koordinatalari deyiladi.

Agar $x=(x_1,x_2,...,x_m)$, $y=(y_1,y_2,...,y_m)$ nuqtalar uchun $x_1=y_1$, $x_2=y_2,..., x_m=y_m$ bo'lsa, $x=y$ deyiladi.

Faraz qilaylik,

$$x=(x_1,x_2,...,x_m), y=(y_1,y_2,...,y_m)$$

lar (1) to'plamning ixtiyoriy ikki nuqtasi bo'lsin. Ushbu

$$\sqrt{\sum_{k=1}^m (y_k - x_k)^2}$$

miqdor x va y nuqtalar orasidagi masofa deyiladi va $\rho(x,y)$ kabi belgilanadi:

$$\rho(x,y)=\sqrt{\sum_{k=1}^m (y_k - x_k)^2} . \quad (2)$$

Endi masofaning xossalari keltiramiz:

1) Har doim $\rho(x,y) \geq 0$ va $\rho(x,y)=0 \Leftrightarrow x=y$ bo'ladi.

◀ (2) munosabatga ko'ra, har doim $\rho(x,y) \geq 0$ bo'ladi. Agar $\rho(x,y)=0$ bo'lsa, unda

$$(y_1 - x_1)^2 + (y_2 - x_2)^2 + ... + (y_m - x_m)^2 = 0$$

bo'lib, natijada $x_1=y_1, x_2=y_2,...,x_m=y_m$, ya'ni $x=y$ bo'lishi kelib chiqadi.

Aksincha, agar $x_1=y_1, x_2=y_2,...,x_m=y_m$ bo'lsa, unda (2) munosabatdan foydalanib $\rho(x,y)=0$ bo'lishini topamiz. ▶

2) $\rho(x,y)$ masofa x va y ularga nisbatan simmetrik bo'ladi:
 $\rho(x,y)=\rho(y,x)$.

◀ Bu xossanig isboti (2) munosabatdan kelib chiqadi:

$$\rho(x, y) = \sqrt{\sum_{k=1}^m (y_k - x_k)^2} = \sqrt{\sum_{k=1}^m (x_k - y_k)^2} = \rho(y, x) \quad \blacktriangleright$$

3) (1) to'plamning ixtiyoriy

$$x = (x_1, x_2, \dots, x_m), \quad y = (y_1, y_2, \dots, y_m), \quad z = (z_1, z_2, \dots, z_m)$$

nuqtalari uchun

$$\rho(x, z) \leq \rho(x, y) + \rho(y, z)$$

tengsizlik o'rinli bo'ladi.

◀ Ma'lumki, ixtiyoriy $a_1, a_2, \dots, a_m$ va $b_1, b_2, \dots, b_m$ haqiqiy sonlar uchun

$$\sqrt{\sum_{k=1}^m (a_k + b_k)^2} \leq \sqrt{\sum_{k=1}^m a_k^2} + \sqrt{\sum_{k=1}^m b_k^2} \quad (3)$$

bo'ladi (qaralsin, [1], 12-bob, 1-§; odatda bu tengsizlikni Koshi-Bunyakovskiy tengsizligi deyiladi). (3) tengsizlikda

$$a_k = y_k - x_k, \quad b_k = z_k - y_k \quad (k = 1, 2, \dots, m)$$

deb topamiz:

$$\sqrt{\sum_{k=1}^m (z_k - x_k)^2} \leq \sqrt{\sum_{k=1}^m (y_k - x_k)^2} + \sqrt{\sum_{k=1}^m (z_k - y_k)^2}.$$

Bu esa

$$\rho(x, z) \leq \rho(x, y) + \rho(y, z)$$

bo'linishi bildiriladi. ▶

Shunday qilib, (1) to'plamda (to'plam elementlari orasida) masofa tushunchasining kiritilishini hamda masofa uchta xossaga ega bo'lishini ko'rdik.

Odatda, (1) to'plam R^m fazo deyiladi. Demak,

$$R^m = \{ (x_1, x_2, \dots, x_m) : x_1 \in R, x_2 \in R, \dots, x_m \in R \}.$$

Endi R^m fazodagi ba'zi bir to'plamlarni keltiramiz.

Aytaylik, biror $a = (a_1, a_2, \dots, a_m) \in R^m$ nuqta va $r > 0$ son berilgan bo'lsin.

Ushbu

$$B_r(a) = \{ (x_1, x_2, \dots, x_m) \in R^m : \sqrt{(x_1 - a_1)^2 + \dots + (x_m - a_m)^2} < r \}$$

qisqacha,

$$B_r(a) = \{ x \in R^m : \rho(x, a) < r \}$$

to'plam markazi a nuqta, radiusi r bo'lgan shar (m o'lchovli shar) deyiladi.

Quyidagi

$$\overline{B}_r(a) = \{ x \in R^m : \rho(x, a) \leq r \}$$

to'plam R^m fazoda yopiq shar,

$$B_r^0(a) = \{ x \in R^m : \rho(x, a) = r \}$$

to'plam esa, R^m fazoda sfera (m o'lchovli sfera) deyiladi.

Ravshanki,

$$\overline{B}_r(a) = B_r(a) \cup B_r^0(a)$$

bo'ladi.

Ushbu

$$\Pi(a_1, \dots, a_m; b_1, b_2, \dots, b_m) = \{ (x_1, x_2, \dots, x_m) \in R^m : a_1 < x_1 < b_1, a_2 < x_2 < b_2, \dots, a_m < x_m < b_m \}$$

to'plam R^m fazoda parallelepiped deyiladi, bunda $a_1, a_2, \dots, a_m; b_1, b_2, \dots, b_m$ haqiqiy sonlar.

2°. R^m fazoda nuqtaning atrofi.

Biror $x^0 = (x_1^0, x_2^0, \dots, x_m^0) \in R^m$ nuqta hamda $\varepsilon > 0$ son berilgan bo'lsin.

1-ta'rif. Markazi x^0 nuqtada radiusi ε bo'lgan R^m fazodagi shar, $x^0 \in R^m$ nuqtaning sferik atrofi deyiladi va $U_\varepsilon(x^0)$ kabi belgilanadi:

$$U_\varepsilon(x^0) = \{x \in R^m : \rho(x, x^0) < \varepsilon\}.$$

2-ta'rif. Ushbu

$$\Pi(\delta_1, \delta_2, \dots, \delta_m) = \{(x_1, x_2, \dots, x_m) \in R^m :$$

$$: x_1^0 - \delta_1 < x_1 < x_1^0 + \delta_1, x_2^0 - \delta_2 < x_2 < x_2^0 + \delta_2, \dots, x_m^0 - \delta_m < x_m < x_m^0 + \delta_m\}$$

parallelepiped x^0 nuqtaning parallelepipedial atrofi deyiladi va $\bar{U}_{\delta_1, \delta_2, \dots, \delta_m}(x^0)$ kabi belgilanadi, bunda $\delta_1 > 0, \delta_2 > 0, \dots, \delta_m > 0$.

R^m fazodagi nuqtaning bu atroflari orasidagi munosabatni quyidagi lemma ifodalaydi.

Lemma. $x^0 \in R^m$ nuqtaning har qanday $U_\varepsilon(x^0)$ sferik atrofi olinganda ham har doim x^0 nuqtaning shunday $\bar{U}_{\delta_1, \delta_2, \dots, \delta_m}(x^0)$ parallelepipedial atrofi topiladiki,

$$\bar{U}_{\delta_1, \delta_2, \dots, \delta_m}(x^0) \subset U_\varepsilon(x^0)$$

bo'ladi.

Shuningdek, x^0 nuqtaning har qanday $\bar{U}_{\delta_1, \delta_2, \dots, \delta_m}(x^0)$ parallelepipedial atrofi olinganda ham har doim x^0 nuqtaning shunday $U_\varepsilon(x^0)$ sferik atrofi topiladiki,

$$\bar{U}_\varepsilon(x^0) \subset \bar{U}_{\delta_1, \delta_2, \dots, \delta_m}(x^0)$$

bo'ladi.

◀ $x^0 \in R^m$ nuqtaning sferik atrofi

$$U_\varepsilon(x^0) = \{x \in R^m : \rho(x, x^0) < \varepsilon\}$$

berilgan bo'lsin. Demak, $\varepsilon > 0$ son berilgan. Unga ko'ra $\delta < \frac{\varepsilon}{\sqrt{m}}$ tengsizlikni

qanoatlantiruvchi δ sonni olib, x^0 nuqtaning ushbu

$$\bar{U}_\delta(x^0) = \bar{U}_{\delta\delta\dots\delta}(x^0) = \{(x_1, x_2, \dots, x_m) \in R^m :$$

$$: x_1^0 - \delta < x_1 < x_1^0 + \delta, x_2^0 - \delta < x_2 < x_2^0 + \delta, \dots, x_m^0 - \delta < x_m < x_m^0 + \delta\}$$

parallelepipedial atrofini tuzamiz. Natijadi x^0 nuqtaning

$$U_\varepsilon(x^0) \supset \bar{U}_\delta(x^0)$$

atroflariga ega bo'lamiz.

Aytaylik, $\forall x \in \bar{U}_\delta(x^0)$ bo'lsin. Unda

$$|x_k - x_k^0| < \delta \quad (k = 1, 2, \dots, m)$$

bo'lib,

$$\sqrt{\sum_{k=1}^m (x_k - x_k^0)^2} < \sqrt{\sum_{k=1}^m \delta^2} = \delta \cdot \sqrt{m}$$

bo'ladi. Yuqoridagi $\delta < \frac{\varepsilon}{\sqrt{m}}$ tengsizlikni e'tiborga olib topamiz:

$$\sqrt{\sum_{k=1}^m (x_k - x_k^0)^2} < \varepsilon$$

Demak, $\rho(x, x^0) < \varepsilon$ bo'lib, $x \in U_\varepsilon(x^0)$ bo'ladi. Bundan

$$\bar{U}_\delta(x^0) \subset U_\varepsilon(x^0)$$

bo'lishi kelib chiqadi.

$x^0 \in R^m$ nuqtaning parallelepipedial atrofi

$$\bar{U}_{\delta_1\delta_2\dots\delta_m}(x^0) = \{(x_1, x_2, \dots, x_m) \in R^m :$$

$$: x_1^0 - \delta_1 < x_1 < x_1^0 + \delta_1, x_2^0 - \delta_2 < x_2 < x_2^0 + \delta_2, \dots, x_m^0 - \delta_m < x_m < x_m^0 + \delta_m \}$$

berilgan bo'lsin. Berilgan $\delta_1, \delta_2, \dots, \delta_m$ musbat sonlar yordamida

$$\varepsilon = \min\{\delta_1, \delta_2, \dots, \delta_m\}$$

sonini topib, x^0 nuqtaning ushbu

$$U_\varepsilon(x^0) = \{x \in R^m : \rho(x, x^0) < \varepsilon\}$$

sferik atrofni tuzamiz. Natijada x^0 nuqtaning

$$U_\varepsilon(x^0) \supset \bar{U}_{\delta_1 \delta_2 \dots \delta_m}(x^0)$$

atroflariga ega bo'lamiz.

Aytaylik, $\forall x \in U_\varepsilon(x^0)$ bo'lsin. U holda

$$\rho(x, x^0) = \sqrt{\sum_{k=1}^m (x_k - x_k^0)^2} < \varepsilon \leq \delta_k \quad (k = 1, 2, \dots, m)$$

bo'lib,

$$|x_k - x_k^0| < \delta_k \quad (k = 1, 2, \dots, m)$$

bo'ladi. Bundan esa $x \in \bar{U}_{\delta_1 \delta_2 \dots \delta_m}(x^0)$ bo'lishi kelib chiqadi. Demak,

$$U_\varepsilon(x^0) \subset \bar{U}_{\delta_1 \delta_2 \dots \delta_m}(x^0). \blacktriangleright$$

Bu lemma R^m fazo nuqtasining bir atrofidan ikkinchi atrofiga o'tishi imkonini beradi.

3°. R^m fazoda ochiq va yopiq to'plamlar.

Aytaylik, R^m fazoda biror G to'plam ($G \subset R^m$) berilgan bo'lib, $x^0 \in G$ bo'lsin.

Agar x^0 nuqta G to'plamga tegishli bo'lgan $U_\varepsilon(x^0)$ atrofga ega bo'lsa, $(U_\varepsilon(x^0) \subset G)$ x^0 nuqta G to'plamning ichki nuqtasi deyiladi.

3-ta'rif G to'plamning har bir nuqtasi uning ichki nuqtasi bo'lsa, u ochiq to'plam deyiladi.

1-misol. R^m fazodagi ushbu

$$B_r(a) = \{x \in R^m : \rho(x, a) < r\}$$

sharning ochiq to'plam ekanligi ko'rsatilsin.

◀ $\forall x^0 \in B_r(a)$ nuqtani olamiz. Unda

$$r - \rho(x^0, a)$$

miqdor musbat bo'ladi. Uni δ deylik: $\delta = r - \rho(x^0, a)$. (22-chizma)

Endi x^0 nuqtaning ushbu

$$U_\delta(x^0) = \{x \in R^m : \rho(x, x^0) < \delta\}$$

atrofini qaraymiz.

Bunda $U_\delta(x^0) \subset B_r(a)$ bo'ladi. Haqiqatdan ham,

$$\forall x \in U_\delta(x^0) \Rightarrow \rho(x, x^0) < \delta$$

bo'lib, masofaning 3)-xossasiga ko'ra

$$\rho(x, a) \leq \rho(x, x^0) + \rho(x^0, a) < \delta + \rho(x^0, a) = r$$

bo'ladi. Demak,

$$\forall x \in U_\delta(x^0) \Rightarrow x \in B_r(a)$$

Bundan $U_\delta(x^0) \subset B_r(a)$ bo'lishi kelib chiqadi.

Demak, $B_r(a)$ to'plamning har bir nuqtasi uning ichki nuqtasi bo'ladi.

Binobarin, $B_r(a)$ ochiq to'plam. ▶

Aytaylik, $F \subset R^m$ to'plam hamda $x^0 \in R^m$ nuqta berilgan bo'lsin. Agar x^0 nuqtaning ixtiyoriy $U_\varepsilon(x^0)$ atrofida ($\forall \varepsilon > 0$) F to'plamning x^0 dan farqli kamida bitta nuqtasi bo'lsa, x^0 nuqta F to'plamning limit nuqtasi deyiladi.

Masalan, ushbu

$$B_r(a) = \{x \in R^m : \rho(x, a) < r\}$$

to'plamning har bir nuqtasi uning limit nuqtasi bo'ladi. Ayni paytda,

$$B_r^0 = \{x \in R^m : \rho(x, a) = r\}$$

to'plamning barcha nuqtalari ham shu $B_r(a)$ to'plamning limit nuqtasi bo'ladi.

Biroq, bu limit nuqtalar $B_r(a)$ to'plamga tegishli bo'lmaydi.

4-ta'rif. Agar $F \subset R^m$ to'plamning barcha limit nuqtalari shu to'plamga tegishli bo'lsa, F yopiq to'plam deyiladi.

Masalan,

$$\overline{B_r}(a) = \{x \in R^m : \rho(x, a) \leq r\}$$

to'plam (R^m fazodagi yopiq shar) yopiq to'plam bo'ladi

Biror $M \subset R^m$ to'plam hamda $x^0 \in R^m$ nuqtani qaraylik.

Agar x^0 nuqtaning ixtiyoriy $U_\varepsilon(x^0)$ atrofida ham M to'plamning, ham $R^m \setminus M$ to'plamning nuqtalari bo'lsa, x^0 nuqta M to'plamning chegaraviy nuqtasi deyiladi. M to'plamning barcha chegaraviy nuqtalari uning chegarasini tashkil etadi. M to'plamning chegarasi $\partial(M)$ kabi belgilanadi.

Masalan,

$$B_r^0(a) = \{x \in R^m : \rho(x, a) = r\}$$

to'plam

$$B_r(a) = \{x \in R^m : \rho(x, a) < r\}$$

to'plamning chegarasi bo'ladi:

$$\partial(B_r(a)) = B_r^0(a).$$

Agar $F \subset R^m$ to'plamning chegarasi $\partial(F)$ shu to'plamga tegishli bo'lsa, F yopiq to'plam bo'ladi.

Masalan,

$$\overline{B_r}(a) = \{x \in R^m : \rho(x, a) \leq r\}$$

yopiq to‘plam bo‘ladi, chunki

$$\partial(\overline{B_r}(a)) = B_r^0(a) \subset \overline{B_r}(a).$$

Mashqlar

1. Agar $G_1 \subset R^m$, $G_2 \subset R^m$ ochiq to‘plamlar bo‘lsa,

$$G_1 \cup R^m, G_2 \cap R^m$$

to‘plamlarning ochiq to‘plam bo‘lishi ko‘rsatilsin.

2. Agar $F_1 \subset R^m$, $F_2 \subset R^m$ yopiq to‘plamlar bo‘lsa,

$$F_1 \cup R^m, F_2 \cap R^m$$

to‘plamlarning yopiq to‘plam bo‘lishi ko‘rsatilsin.

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